# Instalación de DOCKER en Kali Linux

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# INTRODUCCIÓN

El presente documento expone el procedimiento técnico para la implementación del motor de contenedores **Docker** en el sistema operativo **Kali Linux**, haciendo especial énfasis en la resolución de conflictos derivados de la integridad de los repositorios. Durante el despliegue, se identificaron anomalías críticas relacionadas con la caducidad de firmas criptográficas **GPG** y la obsolescencia de índices en los espejos de red, situaciones que impiden la actualización estándar del sistema.

A través de una metodología de diagnóstico y respuesta, esta guía detalla la transición desde un estado de error **404 Not Found** hacia una arquitectura funcional. Se aborda la gestión de llaveros de confianza mediante el uso `de`

`gpg --dearmor`, garantizando la autenticidad de los paquetes descargados.

El objetivo de este material es servir como recurso de consulta para cualquiera (estudiante, profesional o alguien que esta comenzando con Docker en Kali Linux u otros entornos Debian) que esté realizando la tarea de administrar un sistema que requieran mitigar fallos de sincronización en entornos basados en **Debian**, consolidando la documentación técnica como un pilar fundamental en la gestión de infraestructura de TI.

OZONE

# INSTALACIÓN DOCKER EN KALI LINUX

1. Verificar si no está instalado.

   **`docker  --version`**

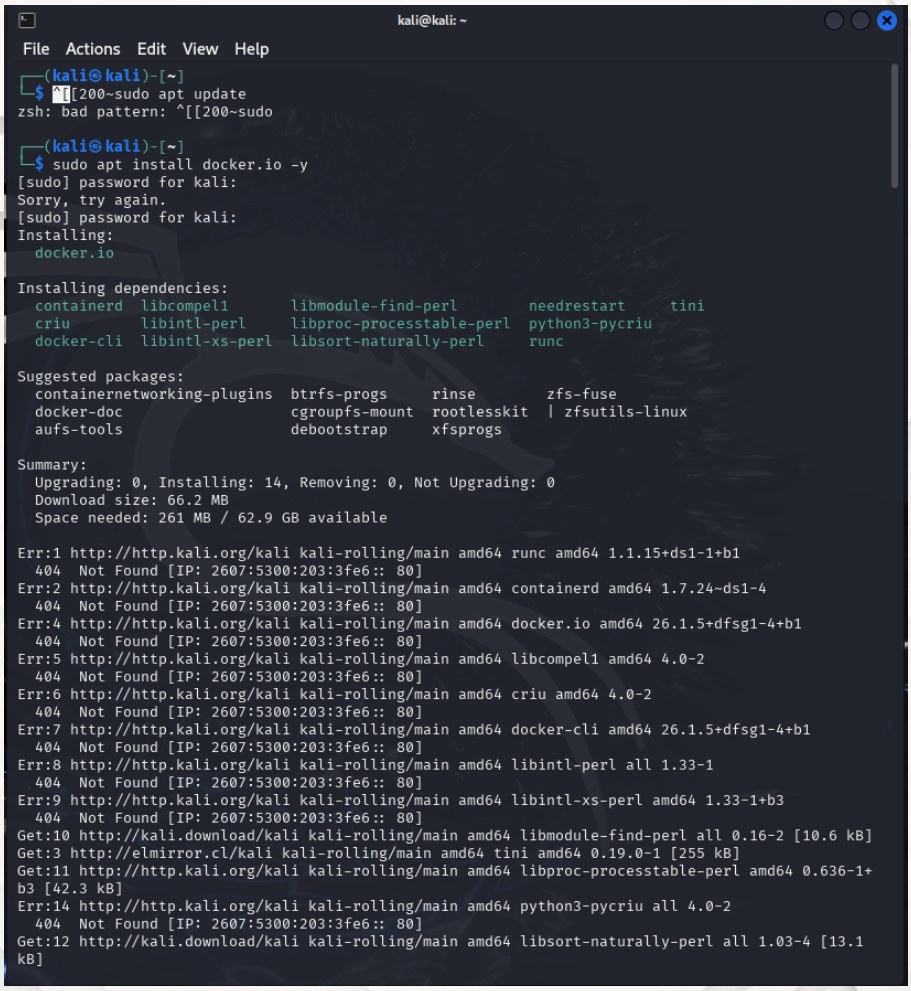
   si no retorna nada, no está instalado.

2. Lanzamos la instalación de Docker

   **`sudo apt update`**

   **`sudo apt install docker.io -y`**

3. En mi caso me dio una lista de fallos y finalmente no se instaló y el porqué de esta documentación por si a alguien más le sucede.



```
kali@kali: ~
File  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ ^[[200~sudo apt update
zsh: bad pattern: ^[[200~sudo

┌──(kali㉿kali)-[~]
└─$ sudo apt install docker.io -y
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Installing:
  docker.io

Installing dependencies:
  containerd  libcompel1       libmodule-find-perl        needrestart     tini
  criu        libintl-perl     libproc-processtable-perl  python3-pycriu
  docker-cli  libintl-xs-perl  libsort-naturally-perl     runc

Suggested packages:
  containernetworking-plugins  btrfs-progs     rinse        zfs-fuse
  docker-doc                   cgroupfs-mount  rootlesskit  | zfsutils-linux
  aufs-tools                   debootstrap     xfsprogs

Summary:
  Upgrading: 0, Installing: 14, Removing: 0, Not Upgrading: 0
  Download size: 66.2 MB
  Space needed: 261 MB / 62.9 GB available

Err:1 http://http.kali.org/kali kali-rolling/main amd64 runc amd64 1.1.15+ds1-1+b1
  404  Not Found [IP: 2607:5300:203:3fe6:: 80]
Err:2 http://http.kali.org/kali kali-rolling/main amd64 containerd amd64 1.7.24~ds1-4
  404  Not Found [IP: 2607:5300:203:3fe6:: 80]
Err:4 http://http.kali.org/kali kali-rolling/main amd64 docker.io amd64 26.1.5+dfsg1-4+b1
  404  Not Found [IP: 2607:5300:203:3fe6:: 80]
Err:5 http://http.kali.org/kali kali-rolling/main amd64 libcompel1 amd64 4.0-2
  404  Not Found [IP: 2607:5300:203:3fe6:: 80]
Err:6 http://http.kali.org/kali kali-rolling/main amd64 criu amd64 4.0-2
  404  Not Found [IP: 2607:5300:203:3fe6:: 80]
Err:7 http://http.kali.org/kali kali-rolling/main amd64 docker-cli amd64 26.1.5+dfsg1-4+b1
  404  Not Found [IP: 2607:5300:203:3fe6:: 80]
Err:8 http://http.kali.org/kali kali-rolling/main amd64 libintl-perl all 1.33-1
  404  Not Found [IP: 2607:5300:203:3fe6:: 80]
Err:9 http://http.kali.org/kali kali-rolling/main amd64 libintl-xs-perl amd64 1.33-1+b3
  404  Not Found [IP: 2607:5300:203:3fe6:: 80]
Get:10 http://kali.download/kali kali-rolling/main amd64 libmodule-find-perl all 0.16-2 [10.6 kB]
Get:3 http://elmirror.cl/kali kali-rolling/main amd64 tini amd64 0.19.0-1 [255 kB]
Get:11 http://http.kali.org/kali kali-rolling/main amd64 libproc-processtable-perl amd64 0.636-1+
b3 [42.3 kB]
Err:14 http://http.kali.org/kali kali-rolling/main amd64 python3-pycriu all 4.0-2
  404  Not Found [IP: 2607:5300:203:3fe6:: 80]
Get:12 http://kali.download/kali kali-rolling/main amd64 libsort-naturally-perl all 1.03-4 [13.1
kB]
```

4. Lo que está ocurriendo es **un error común cuando los paquetes cambian de versión en los repositorios oficiales**, pero mi sistema aún está apuntando a versiones antiguas que ya no existen (por eso da error 404). Entonces nos lanzamos a actualizar la lista de paquetes con:

   **`sudo apt update`**

   que para mi sorpresa me dio error también.

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update

Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]
Err:1 http://kali.download/kali kali-rolling InRelease
  The following signatures couldn't be verified because the public key is not available: NO_PUBKE
Y ED[redacted]
All packages are up to date.
Warning: An error occurred during the signature verification. The repository is not updated and t
he previous index files will be used. GPG error: http://kali.download/kali kali-rolling InRelease
: The following signatures couldn't be verified because the public key is not available: NO_PUBKE
Y ED[redacted]
Warning: Failed to fetch http://http.kali.org/kali/dists/kali-rolling/InRelease  The following si
gnatures couldn't be verified because the public key is not available: NO_PUBKEY ED[redacted]
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
```

El error en este punto, básicamente se trata de que el sistema **no puede verificar la autenticidad de los paquetes del repositorio oficial de Kali**. Eso bloquea la instalación de Docker (y de cualquier otro paquete que dependa de ese repositorio).

**¿Qué significa el error?**

Kali Linux necesita una **clave GPG pública** para verificar que los paquetes que descarga del repositorio son auténticos.

El error:

**NO_PUBKEY EDxxxxxxxxxxxxxxxx**

Indica que esa clave no esta en el sistema, entonces la solución pasa por añadir la clave faltante de forma manual, ejecutando la siguiente línea en la terminal para descargar e importar la clave publica correcta.

**sudo apt-key adv --keyserver keyserver.ubuntu.com --recv-keys EDxxxxxxxxxxxxxxx**

cómo vuelve a dar error**,** (**apt-key** va en desuso) usamos este otro método:

**curl -fsSL https://archive.kali.org/archive-key.asc | sudo gpg --dearmor -o /usr/share/keyrings/kali-archive-keyring.gpg**

preguntará si deseamos sobrescribir y decimos que **SI.**

Ahora si podemos volver al paso 2,

**sudo apt update.**

```
┌──(kali㉿kali)-[~]
└─$ sudo apt-key adv --keyserver keyserver.ubuntu.com --recv-keys ED[redacted]

sudo: apt-key: command not found

┌──(kali㉿kali)-[~]
└─$ curl -fsSL https://archive.kali.org/archive-key.asc | sudo gpg --dearmor -o /usr/share/keyrin
gs/kali-archive-keyring.gpg

File '/usr/share/keyrings/kali-archive-keyring.gpg' exists. Overwrite? (y/N) y

┌──(kali㉿kali)-[~]
└─$ sudo apt update

Get:1 http://elmirror.cl/kali kali-rolling InRelease [41.5 kB]
Get:2 http://elmirror.cl/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://elmirror.cl/kali kali-rolling/main amd64 Contents (deb) [52.0 MB]
Get:4 http://elmirror.cl/kali kali-rolling/contrib amd64 Packages [121 kB]
Get:5 http://elmirror.cl/kali kali-rolling/contrib amd64 Contents (deb) [327 kB]
Get:6 http://elmirror.cl/kali kali-rolling/non-free amd64 Packages [204 kB]
Get:7 http://elmirror.cl/kali kali-rolling/non-free amd64 Contents (deb) [914 kB]
Get:8 http://elmirror.cl/kali kali-rolling/non-free-firmware amd64 Packages [10.6 kB]
Get:9 http://elmirror.cl/kali kali-rolling/non-free-firmware amd64 Contents (deb) [24.3 kB]
Fetched 74.6 MB in 7s (10.7 MB/s)
1640 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

5. A continuación la segunda parte del paso 2,

```
sudo apt install docker.io -y
```

y comenzará la instalación de Docker.

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install docker.io -y

Installing:
  docker.io

Installing dependencies:
  containerd     docker-cli    libintl-xs-perl            libsort-naturally-perl  runc
  criu           libcompel1    libmodule-find-perl        needrestart             tini
  docker-buildx  libintl-perl  libproc-processtable-perl  python3-pycriu

Suggested packages:
  containernetworking-plugins  btrfs-progs     rinse        zfs-fuse
  docker-doc                   cgroupfs-mount  rootlesskit  | zfsutils-linux
  aufs-tools                   debootstrap     xfsprogs

Summary:
  Upgrading: 0, Installing: 15, Removing: 0, Not Upgrading: 1640
  Download size: 81.1 MB / 81.4 MB
  Space needed: 335 MB / 62.9 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 runc amd64 1.1.15+ds1-2+b3 [3,229 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 libcompel1 amd64 4.1-1 [64.0 kB]
Get:5 http://kali.download/kali kali-rolling/main amd64 criu amd64 4.1-1 [560 kB]
Get:6 http://http.kali.org/kali kali-rolling/main amd64 docker-buildx amd64 0.13.1+ds1-3 [13.2 MB
]
Get:7 http://http.kali.org/kali kali-rolling/main amd64 docker-cli amd64 26.1.5+dfsg1-9+b3 [7,332
 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 containerd amd64 1.7.24~ds1-5+b1 [32.8 MB
]
Get:8 http://kali.download/kali kali-rolling/main amd64 libintl-perl all 1.35-1 [690 kB]
Get:9 http://kali.download/kali kali-rolling/main amd64 libintl-xs-perl amd64 1.35-1 [15.3 kB]
Get:10 http://kali.download/kali kali-rolling/main amd64 libmodule-find-perl all 0.17-1 [10.7 kB]
Get:12 http://kali.download/kali kali-rolling/main amd64 python3-pycriu all 4.1-1 [43.7 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 docker.io amd64 26.1.5+dfsg1-9+b3 [23.0 M
B]
Get:11 http://mirror.ufro.cl/kali kali-rolling/main amd64 needrestart all 3.11-1 [68.6 kB]
Fetched 81.1 MB in 2s (34.4 MB/s)
Selecting previously unselected package runc.
(Reading database ... 406689 files and directories currently installed.)
Preparing to unpack .../00-runc_1.1.15+ds1-2+b3_amd64.deb ...
Unpacking runc (1.1.15+ds1-2+b3) ...
Selecting previously unselected package containerd.
Preparing to unpack .../01-containerd_1.7.24~ds1-5+b1_amd64.deb ...
Unpacking containerd (1.7.24~ds1-5+b1) ...
Selecting previously unselected package tini.
Preparing to unpack .../02-tini_0.19.0-1_amd64.deb ...
Unpacking tini (0.19.0-1) ...
```

6. Una vez finalizada la instalación se recomienda volver al paso 1 que ahora si da la información correcta

```
┌──(kali㉿kali)-[~]
└─$ docker --version
Docker version 26.1.5+dfsg1, build a72d7cd
```

7. Verificamos que el servicio se encuentra activo con

**`sudo systemctl status Docker`**

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl status docker

● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Thu 2025-05-08 18:46:39 EDT; 2min 16s ago
 Invocation: 572[redacted]
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 5157 (dockerd)
      Tasks: 10
     Memory: 34.6M (peak: 36.5M)
        CPU: 263ms
     CGroup: /system.slice/docker.service
             └─5157 /usr/sbin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

May 08 18:46:38 kali systemd[1]: Starting docker.service - Docker Application Container Engine...
May 08 18:46:38 kali (dockerd)[5157]: docker.service: Referenced but unset environment variable >
May 08 18:46:39 kali dockerd[5157]: time="2025-05-08T18:46:39.017367861-04:00" level=info msg="S>
May 08 18:46:39 kali dockerd[5157]: time="2025-05-08T18:46:39.126572079-04:00" level=info msg="L>
May 08 18:46:39 kali dockerd[5157]: time="2025-05-08T18:46:39.411373314-04:00" level=info msg="L>
May 08 18:46:39 kali dockerd[5157]: time="2025-05-08T18:46:39.425175898-04:00" level=info msg="D>
May 08 18:46:39 kali dockerd[5157]: time="2025-05-08T18:46:39.425263098-04:00" level=info msg="D>
May 08 18:46:39 kali dockerd[5157]: time="2025-05-08T18:46:39.448720765-04:00" level=info msg="A>
May 08 18:46:39 kali systemd[1]: Started docker.service - Docker Application Container Engine.
```

En la que vemos el indicativo de active marcado en “azul vincapervinca” (Snatch)

8. Para estar más seguros realizamos una prueba rápida con el contenedor de prueba oficial de Docker.

```
sudo docker run hello-world
```

Si te saludan de vuelta con un **"Hello from Docker!"** entonces todo debería estar ok.

```
┌──(kali㉿kali)-[~]
└─$ sudo docker run hello-world

Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
e6590344b1a5: Pull complete
Digest: sha256:[redacted]
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

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